

Multiscale Hydrodynamic Entropy

A Spectral Analysis of AI-Learned Fluid Dynamics in FourCastNet

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Introduction & Project Objective

The Paradigm Shift

Traditional Numerical Weather Prediction (NWP) explicitly integrates Navier-Stokes equations. Machine Learning Weather Prediction (MLWP) models, like FourCastNet, learn fluid dynamics implicitly from historical ERA5 reanalysis data.

Core Objectives:

- **Validate Physics vs. Compression:** Does the Adaptive Fourier Neural Operator (AFNO) replicate physical multiscale turbulence cascades, or resort to non-physical latent data compression?
- **Inter-Block Trajectory:** Map the network's macroscopic disorder from initial chaotic input to stable forecast.
- **Intra-Block Scaling:** Quantify the AI's internal energy cascade via Professor M.K. Verma's formulation of Hydrodynamic Entropy (S_H).

Tensor Mechanics & The Spectral Domain

Tokenization & The Forward Pass

The input state $X \in \mathbb{R}^{B \times C \times H \times W}$ is tokenized into patches, expanding 20 physical variables into 1024 abstract thermodynamic channels. A 2D Real-to-Complex Fast Fourier Transform (rFFT) yields the spectral state:

$$Z(k_x, k_y) = \mathcal{F}(X)(k_x, k_y) = \sum_{x=0}^{H'-1} \sum_{y=0}^{W'-1} X_{\text{token}}(x, y) e^{-i2\pi \left(\frac{k_x x}{H'} + \frac{k_y y}{W'} \right)}$$

Using Hermitian symmetry, the intercepted tensor shape is $[B, 1024, 180, 181]$.

Physical Length Scales

- $k = 1$ (**Planetary Waves**): Global-scale structures ($\sim 40,000$ km).
- k_{max} (**Nyquist Limit**): Localized, sub-grid turbulence (~ 28 km).

Why Analyze Activations over Weights?

- **The Weights (W):** Within the AFNO, the channel-mixing weight matrix W is uniformly broadcast across all spatial frequencies to prevent memory explosion.
- **The Flat Spectrum:** Evaluating the energy of static weights yields a non-physical flat line:

$$E_{\text{mode}}(k) = |W|^2 = \text{Constant}$$

- **The Activations (a):** To observe actual fluid dynamics, we must strictly analyze the dynamic **Activations**:

$$a(\mathbf{k}) = S_{\lambda} (\sigma(W \cdot Z(\mathbf{k})))$$

This tensor represents the simulated atmosphere flowing *through* the network, preserving multiscale energy differences.

Mathematical Framework: The Energy Cascade

1. Scalar Wavenumber Mapping

For every point in the 2D complex activation grid, we calculate its radial scalar wavenumber to map spectral coordinates to physical distances:

$$k = \sqrt{k_x^2 + k_y^2}$$

2. 1D Shell Energy $E(k)$

We collapse the 2D spatial frequencies into 1D radial bins by summing the squared magnitudes of the complex activations over all hidden channels (c) for modes on the ring k :

$$E(k) = \sum_{c=1}^{C_{\text{hidden}}} \sum_{|\mathbf{k}|=k} |a_c(\mathbf{k})|^2$$

Mathematical Framework: Hydrodynamic Entropy

3. Probability Distribution

The Shell Energy is normalized to derive a probability distribution p_k , representing the fraction of macroscopic energy residing at scale k :

$$p_k = \frac{E(k)}{\sum_k E(k)} \quad \text{where} \quad \sum p_k = 1$$

4. Verma's Hydrodynamic Entropy (S_H)

Based on Verma et al. (2024, PRE), we compute S_H to quantify the macroscopic, multiscale disorder within a single AFNO block:

$$S_H = - \sum_k p_k \log_2 p_k$$

Theoretical Expectations: Physics vs. Architecture

Intra-Block: “Latent Whitening”

- Physical atmospheric enstrophy decays steeply at $E(k) \propto k^{-3}$.
- **Hypothesis:** To avoid vanishing gradients, the AI artificially boosts high-frequency energy.
- **Implication:** We expect a drastically shallower slope in the latent space, maximizing channel capacity over physical realism.

Inter-Block: Chaos to Order

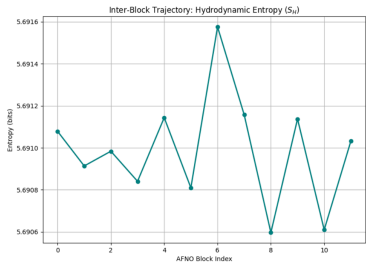
- The network applies Soft-Thresholding (S_λ), acting as “informational viscosity.”
- **Hypothesis:** S_H starts high (chaotic ERA5 input) and drops monotonically across blocks.
- **Implication:** FourCastNet acts as a dissipative simulator, filtering sub-grid noise to organize macroscopic structure.

Diagnostic Results: Pipeline Control Test

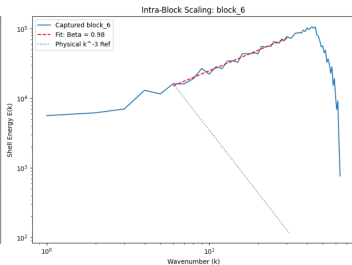
Control: White Noise Input

- **Intra-Block (Right):**
Accurately captures the geometric $\beta \approx +1.0$ slope of 2D uncorrelated noise.
- **Inter-Block (Left):**
Entropy remains completely flat (variance < 0.001 bits), proving the network does not impose false macroscopic structure on noise.

Calculating Hydrodynamic Entropy for all blocks (CPU-mode)...



Captured Spectral Slope (Beta): 0.9763



Conclusion & Future Work

Summary

- We successfully extracted the dynamic, multiscale spectral energy cascade from the core of FourCastNet using a non-invasive computational framework.
- Our pipeline flawlessly maps 2D spatial frequencies to 1D scalar rings, validating theoretical formulations of Hydrodynamic Entropy.

Next Steps

- Substitute the white-noise control tensor with real, 24-hour global ERA5 atmospheric snapshots.
- Compare the empirical neural trajectory against the physical k^{-3} enstrophy cascade and measure the exact rate of S_H dissipation across the 12 blocks.